

Project 11: Model Optimization Experiment Lab

Inference Quantization & Hardware Acceleration Benchmarking

1. Executive Overview

The Model Optimization Lab benchmarks post-training quantization (PTQ) techniques. It evaluates FP16 (Baseline), INT8 (8-bit Quantization), and INT4 (4-bit AWQ) precisions, comparing VRAM footprint (-71.9% memory savings), TTFT latency, throughput (88.2 tok/s), perplexity shift, and quality retention scores.

2. System Architecture & Workflow Pipeline

- Precision Evaluator: FP16 Baseline vs INT8 vs INT4 AWQ
- Hardware Metrics: VRAM Footprint, TTFT Latency, Throughput (tok/s)
- Quality Retention Analyzer: Perplexity shift & Quality Score %
- Gemini 2.5 Flash Trade-Off Engineering Recommendation Engine

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Model Optimization Lab web UI
GET/POST	/api/run-optimization	Runs precision benchmark suite and returns trade-off report

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 11_model_optimization/src && python -m uvicorn server:app --host 127.0.0.1 --port 8011
```

Automated Test Suite Execution:

```
python -m pytest 11_model_optimization/tests/
```